TCP/IP & OSI MODELS

Networking Fundamentals • Cybersecurity

Student Name	Nethra M
Assignment Code	DAY04-TCPIP-OSI
Total Marks	20 marks
Submission Format	PDF report

PART A — OSI / TCP-IP THEORY

1. What is a network protocol?

A network protocol is a set of rules that devices follow to communicate. It defines how data is formatted, addressed, transmitted, received, and checked. Examples include HTTP, TCP, IP, and DNS.

2. What is the OSI Model?

The Open Systems Interconnection (OSI) Model is a seven-layer reference framework that explains how data moves between networked devices. Each layer performs specific functions and works with the layers above and below it.

3. Name all seven OSI layers in correct order.

From Layer 7 to Layer 1: Application, Presentation, Session, Transport, Network, Data Link, and Physical.

4. What is the purpose of the Application Layer?

Layer 7 provides network services to applications, enabling functions such as web browsing, email, and name resolution. Examples of application-layer protocols include HTTP, HTTPS, SMTP, and DNS.

5. What is the purpose of the Transport Layer?

Layer 4 provides communication between applications on different hosts. TCP supports reliable, ordered delivery; UDP provides connectionless datagrams with lower overhead. Port numbers identify application services.

6. What is the purpose of the Network Layer?

Layer 3 provides logical addressing and routing between networks. It uses IP addresses to move packets toward their destination; routers primarily make forwarding decisions at this layer.

7. What is the purpose of the Data Link Layer?

Layer 2 transfers frames over a local network link. It uses MAC addresses for local delivery and may handle media access and error detection. Ethernet and Wi-Fi include Data Link functions.

8. What is the purpose of the Physical Layer?

Layer 1 carries raw bits as electrical, optical, or radio signals. It defines characteristics such as cabling, connectors, radio frequencies, signaling, and data rates.

9. What is the TCP/IP Model?

The TCP/IP Model describes the protocol architecture used by the Internet. A common four-layer version has Application, Transport, Internet, and Network Access/Link layers.

10. Explain the difference between OSI and TCP/IP models.

OSI is a seven-layer reference model useful for learning and troubleshooting. TCP/IP is a four-layer model based on the Internet protocol suite. TCP/IP combines the OSI Presentation and Session functions into its Application layer and generally groups OSI Data Link and Physical functions into Network Access/Link.

11. What is encapsulation?

Encapsulation is the process of adding protocol information as data moves down the network stack for transmission. Transport information is added to application data, an IP header is added to form a packet, and a link-layer header and trailer are added to form a frame.

12. What is decapsulation?

Layer	Name	Purpose / examples
7	Application	Services for applications; HTTP, HTTPS, DNS, SMTP
6	Presentation	Data representation, encryption, compression
5	Session	Dialog and session coordination
4	Transport	TCP/UDP, ports, end-to-end delivery
3	Network	IP addressing and routing

2	Data Link	Frames, MAC addresses, local links

Decapsulation is the reverse process at the receiving device. Each layer reads and removes its relevant header or trailer, then passes the remaining data up to the next layer.

13. Which layer uses IP addresses?

The Network Layer (OSI Layer 3), called the Internet layer in TCP/IP, uses IP addresses for logical addressing and routing.

14. Which layer uses MAC addresses?

The Data Link Layer (OSI Layer 2) uses MAC addresses to deliver frames across a local network segment.

15. Which layer uses TCP and UDP?

The Transport Layer (OSI Layer 4) uses TCP and UDP. TCP provides reliable, ordered delivery, while UDP sends datagrams without TCP's connection and reliability mechanisms.

16. Which layer includes HTTP and HTTPS?

HTTP and HTTPS are Application Layer protocols (OSI Layer 7 and the TCP/IP Application layer). HTTPS protects HTTP communication using TLS.

17. Why should cybersecurity professionals understand networking layers?

Understanding layers helps cybersecurity professionals analyze packet captures, troubleshoot connectivity, investigate incidents, interpret firewall rules, and recognize whether a problem or attack involves an application, transport port, IP route, local link, or physical connection.

18. Explain how network data travels from a browser to a server.

The browser resolves the server name through DNS when needed, then establishes a connection (commonly TCP, or QUIC over UDP for HTTP/3). For HTTPS, TLS protects the HTTP exchange. Data is encapsulated into transport segments or datagrams, IP packets, and link-layer frames. Routers forward packets using IP addresses, while link-layer addresses apply to each local hop. The server decapsulates and processes the request, then sends a response back. The browser receives the response and renders the page.

Quick reference — OSI layers

Layer	Name	Purpose / examples
1	Physical	Signals, bits, cables, radio

PART B — PRACTICAL

Open Windows Command Prompt (cmd) and run the commands below. Record results exactly as displayed on your computer. Results can vary by location, DNS response, and network conditions.

19. Resolved IP address for google.com (nslookup)

Command:

```
nslookup google.com
```

Record one or more returned IPv4/IPv6 addresses. Do not use a guessed or copied address.

20. Number of traceroute hops to google.com

Command:

```
tracert google.com
```

Count the numbered hop lines in the completed trace. A line containing “* * * Request timed out” still represents a hop, although no response was received. Record the hop count shown by your own trace.

```
C:\Users\Hp>tracert google.com

Tracing route to google.com [192.178.173.102]
over a maximum of 30 hops:

  1    13 ms     2 ms     2 ms  10.234.13.145
  2    46 ms    26 ms    40 ms  10.175.5.252
  3     *        *        *     Request timed out.
  4    78 ms   151 ms   148 ms  192.168.174.208
  5   110 ms    76 ms    74 ms  192.168.203.1
  6   111 ms    78 ms    68 ms  192.168.203.2
  7   108 ms    81 ms    66 ms  122.15.231.18
  8   121 ms    60 ms    78 ms  182.19.108.191
  9   104 ms    72 ms    75 ms  74.125.146.82
 10   117 ms    96 ms    79 ms  lcbome-in-f102.1e100.net [192.178.173.102]

Trace complete.

C:\Users\Hp>nslookup google.com
Server:  UnKnown
Address:  10.234.13.145

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4000:100c::8a
          2404:6800:4000:100c::71
          2404:6800:4000:100c::8b
          2404:6800:4000:100c::66
          192.178.173.138
          192.178.173.100
          192.178.173.113
          192.178.173.139
          192.178.173.101
          192.178.173.102


C:\Users\Hp>
```

```
   Autoconfiguration Enabled . . . . : Yes

C:\Users\Hp>ping google.com

Pinging google.com [192.178.173.102] with 32 bytes of data:
Reply from 192.178.173.102: bytes=32 time=82ms TTL=108
Reply from 192.178.173.102: bytes=32 time=136ms TTL=108
Reply from 192.178.173.102: bytes=32 time=105ms TTL=108
Reply from 192.178.173.102: bytes=32 time=72ms TTL=108

Ping statistics for 192.178.173.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 72ms, Maximum = 136ms, Average = 98ms

C:\Users\Hp>tracert google.com

Tracing route to google.com [192.178.173.102]
over a maximum of 30 hops:

  1    13 ms     2 ms     2 ms  10.234.13.145
  2    46 ms    26 ms    40 ms  10.175.5.252
  3     *        *        *     Request timed out.
  4    78 ms   151 ms   148 ms  192.168.174.208
  5   110 ms    76 ms    74 ms  192.168.203.1
  6   111 ms    78 ms    68 ms  192.168.203.2
  7   108 ms    81 ms    66 ms  122.15.231.18
  8   121 ms    60 ms    78 ms  182.19.108.191
  9   104 ms    72 ms    75 ms  74.125.146.82
 10   117 ms    96 ms    79 ms  lcbome-in-f102.1e100.net [192.178.173.102]

Trace complete.

C:\Users\Hp>nslookup google.com
Server:  UnKnown
Address:  10.234.13.145

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4000:100c::8a
          2404:6800:4000:100c::71
          2404:6800:4000:100c::8b
          2404:6800:4000:100c::66
```